LESSON 01 / 05

Definitions and the two formulas

A **sequence** is an ordered list $a_1, a_2, \ldots$; a **series** is a sum of its terms. S_n always means $a_1 + \cdots + a_n$. A sequence is an **arithmetic progression (AP)** when consecutive differences are constant: $a_{n+1} - a_n = d$ for all n.

SHORT DERIVATION

The AP formulas from pairing the ends

$a_n = a + (n-1)d$ by adding d exactly $n-1$ times. For the sum, write S_n forwards and backwards and add: each of the n column pairs totals $a + \ell$ (first + last), so

$$S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + \ell)$$

The second form is "number of terms × average of first and last" — and it is also $n \times$ (middle term) when n is odd.

WORKED EXAMPLE

Worked — AP basics

AP $2, 5, 8, \ldots$: $a_{20} = 2 + 19 \cdot 3 = 59$; $S_{20} = \frac{20}{2}(2 + 59) = 610$. Sum of first 20 positive even integers: $2 \cdot \frac{20 \cdot 21}{2} = 420$.

Means: three numbers in AP iff the middle is the average: $2b = a + c$. Insert 3 arithmetic means between 2 and 14: $d = \frac{14-2}{3+1} = 3$, means $5, 8, 11$.

KEY RESULT

AP facts that win marks

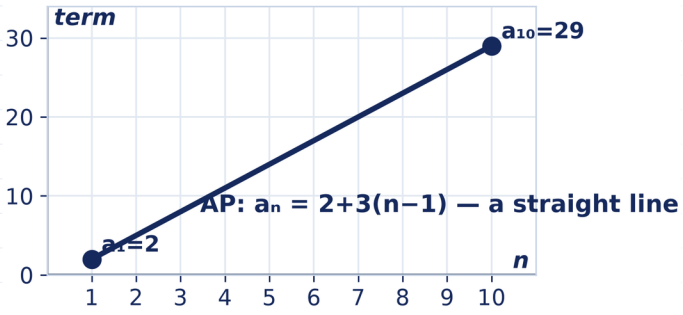
- $a_m + a_n = a_p + a_q$ whenever $m + n = p + q$ (indices trade against each other). In particular terms equidistant from the ends sum to $a + \ell$.
- S_n is a quadratic in n with zero constant term: $S_n = \frac{d}{2}n^2 + (a - \frac{d}{2})n$. Conversely any such quadratic is an AP sum, recovered by $a_n = S_n - S_{n-1}$.
- Three terms in AP: write $a - d, a, a + d$; four terms: $a - 3d, a - d, a + d, a + 3d$ (common difference $2d$) — symmetric parametrisation kills the linear term.
- If a, b, c in AP then b is the **arithmetic mean** of a, c: $A = \frac{a+c}{2}$.

COMMON TRAP

Common traps — AP

- Inserting k means creates $k + 1$ gaps: $d = \frac{b-a}{k+1}$, not $\frac{b-a}{k}$.
- $S_n = \frac{n}{2}(a + \ell)$ needs ℓ to be genuinely the n-th term — confirm the term count first ("sum of terms from 5th to 20th" has 16 terms).
- Given S_n as a formula, $a_1 = S_1$, not "$S_n - S_{n-1}$ at $n = 1$".

An AP plotted against n is linear: slope = d



LESSON 02 / 05

Definitions and the shift-and-subtract proof

A **geometric progression (GP)** has constant ratio $\frac{a_{n+1}}{a_n} = r$ (terms assumed nonzero for the ratio definition). Hence $a_n = ar^{n-1}$.

SHORT DERIVATION

The GP sum — the trick the whole chapter reuses

Write $S_n = a + ar + \cdots + ar^{n-1}$ and $rS_n = ar + \cdots + ar^n$. Subtract: everything telescopes except the ends, $(1-r)S_n = a(1-r^n)$, so for $r \neq 1$

$$S_n = \frac{a(r^n-1)}{r-1} = \frac{a(1-r^n)}{1-r}$$

(both forms identical; pick the one avoiding negatives). If $|r|<1$ then $r^n \to 0$ and

$$S_\infty = \frac{a}{1-r} \qquad (|r|<1 \text{ only})$$

WORKED EXAMPLE

Worked — GP basics

GP $3, 6, 12, \ldots$: $a_{10} = 3\cdot 2^9 = 1536$; $S_{10} = 3(2^{10}-1) = 3069$. Infinite: $1 + \frac{1}{2} + \frac{1}{4} + \cdots = \frac{1}{1-1/2} = 2$ (partial sum after 10 terms 1.9980 ✓).

Means: insert 3 geometric means between 1 and 81: $r = \left(\frac{81}{1}\right)^{1/4} = 3$, means $3, 9, 27$. Geometric mean of $2, 8, 32$ is $(2\cdot 8\cdot 32)^{1/3} = 8$.

KEY RESULT

GP facts that win marks

- $a_m a_n = a_p a_q$ whenever $m+n = p+q$; terms equidistant from the ends have constant product $a\cdot\ell$.
- Three terms in GP: $\frac{a}{r}, a, ar$; four terms: $\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$.
- Every term (except possibly signs) satisfies $a_n^2 = a_{n-1}a_{n+1}$: a, b, c in GP iff $b^2 = ac$.
- If each term of a GP is multiplied by a constant, raised to a power, or reciprocated, the result is still a GP (ratio r, r^k, $\frac{1}{r}$ respectively). Products/quotients of two GPs are GPs.
- A GP with $r>0$ has all terms of one sign; $r<0$ alternates. Log of a positive GP is an AP — the bridge between the two progressions.

KEY RESULT

The geometric mean and AM–GM (two positive numbers)

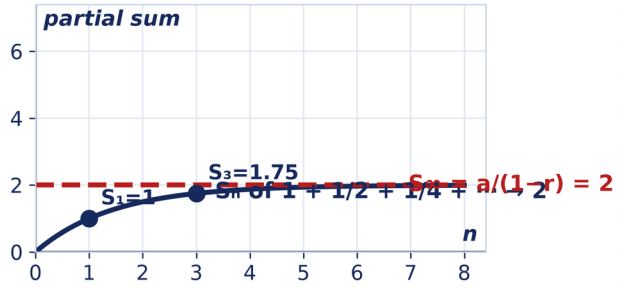
$G = \sqrt{ab}$ is defined by a, G, b in GP. For $a, b > 0$: $\frac{a+b}{2} \geq \sqrt{ab}$, equality iff $a = b$ — because it rearranges to $(\sqrt{a}-\sqrt{b})^2 \geq 0$. AM–GM is examined in depth on sheet 03.

COMMON TRAP

Common traps — GP

- S_∞ used when $|r| \geq 1$: the partial sums do not converge; there is no finite sum. At $r=1$, $S_n = na$; at $r=-1$ the sums oscillate.
- Sign chaos from mixing $\frac{a(r^n-1)}{r-1}$ and $\frac{a(1-r^n)}{1-r}$ mid-line.
- Inserting k geometric means: the exponent is $\frac{1}{k+1}$, and for even roots with negative $\frac{b}{a}$ there may be no real r.
- $a_n = ar^{n-1}$: the exponent is $n-1$, not n.

Infinite GP: partial sums approach a/(1−r) when |r|<1



LESSON 03 / 05

The means and the master inequality

KEY RESULT

AM ≥ GM, with equality condition

For positive reals, $\frac{a+b}{2} \geq \sqrt{ab}$, equality iff $a = b$; for n positive numbers, $\frac{a_1+\cdots+a_n}{n} \geq (a_1\cdots a_n)^{1/n}$, equality iff all are equal. The hypothesis **positive** is not decoration: AM–GM is false for negative numbers.

WORKED EXAMPLE

Worked — AM-GM as an optimiser

Least value of $x + \frac{4}{x}$ for $x > 0$: AM-GM gives $\geq 2\sqrt{x \cdot \frac{4}{x}} = 4$, attained when $x = \frac{4}{x}$, i.e. $x = 2$ ✓. The recipe: force the product to be constant, apply AM-GM, then **check the equality case is attainable** in the domain. If equality needs $x = 2$ but the domain is $x \geq 3$, the bound is not the minimum — the function is increasing there and the answer is at $x = 3$ (value $\frac{13}{3}$). Unattainable equality is the classic trap.

The special sums (official syllabus)

SHORT DERIVATION

Where the power sums come from — one telescoping identity

$(k+1)^{m+1} - k^{m+1}$ expands with leading terms $(m+1)k^m + \cdots$; summing over $k = 1..n$ telescopes the left side to $(n+1)^{m+1} - 1$ and yields $\sum k^m$ in terms of lower sums. With $m = 1$: $(k+1)^2 - k^2 = 2k + 1$ gives $\sum k = \frac{n(n+1)}{2}$; iterating gives the rest. You can re-derive any of them in the exam if forgotten.

KEY RESULT

Memorise these three (the fourth is a bonus)

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$$

The cube sum is the **square** of the first sum — remember it as a picture, not a formula. (Bonus, beyond the syllabus list: $\sum k^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$.)

WORKED EXAMPLE

Worked — special sums

$\sum_{1}^{10} k^2 = \frac{10 \cdot 11 \cdot 21}{6} = 385$; $\sum_{1}^{10} k^3 = 55^2 = 3025$.

Mixed: $\sum_{k=1}^{n} (3k^2 - 2k) = 3 \cdot \frac{n(n+1)(2n+1)}{6} - 2 \cdot \frac{n(n+1)}{2}$; at $n = 10$: $1155 - 110 = 1045$ (direct check by script in math_checks). Sums of odd/even terms: first n odd numbers sum to n^2 (AP or $2\sum k - n$).

COMMON TRAP

Common traps — means and sums

- AM-GM applied to quantities that can be negative, or with equality case outside the domain.
- "n-term AM-GM" confused with repeated two-term AM-GM giving a weaker bound whose equality case is inconsistent.
- $\sum k^3$ written as $\frac{n(n+1)}{2}$ (forgotten square).
- Sums indexed from $k = 0$ or shifted ranges substituted into formulas built for $k = 1..n$: adjust by subtracting the missing head.

INTUITION

Reading a sum before computing it

Ask: is the general term a polynomial in k? Then split into power sums. A product of polynomial × exponential? AGP (sheet 04). A rational function with AP-spaced factors in the denominator? Telescope (sheet 04). Anything else, suspect a disguised difference $f(k) - f(k+1)$.

LESSON 04 / 05

ASSUMPTIONS & CONVENTIONS

STATUS — core technique, extension label

The official syllabus names AP, GP, AM/GM and $\sum k, \sum k^2, \sum k^3$. AGP sums and the method of differences are **not separate syllabus lines** — they are one-step derivations from the GP sum formula, and JEE Advanced asks them exactly in that derivable form. Master the derivations, not just the results.

Engine 1: shift-and-subtract (AGP)

An **arithmetico-geometric progression** has terms $(a+kd)r^k$: an AP factor times a GP factor.

SHORT DERIVATION

The AGP sum

Let $S=\sum_{k=0}^{n-1}(a+kd)r^k$. Form $S-rS$: the AP factor's constancy turns the difference into a plain GP plus one leftover term, giving (for $r\neq 1$) a closed form; letting $n\to\infty$ with $|r|<1$,

$$\sum_{k=0}^{\infty}(a+kd)r^k=\frac{a}{1-r}+\frac{dr}{(1-r)^2}$$

Most used case: $\sum_{k=1}^{\infty}kx^k=\frac{x}{(1-x)^2}$ $(|x|<1)$.

Applying shift-subtract twice: $\sum k^2x^k=\frac{x(1+x)}{(1-x)^3}$.

WORKED EXAMPLE

Worked — AGP

$\sum_{k=1}^{\infty}\frac{k}{2^k}=\frac{1/2}{(1-1/2)^2}=2$. Finite check: $1+2x+3x^2+4x^3+5x^4$ at $x=\frac{1}{2}$ is $1+1+\frac{3}{4}+\frac{1}{2}+\frac{5}{16}=\frac{57}{16}$, matching the finite formula $\frac{1-(n+1)x^n+nx^{n+1}}{(1-x)^2}$ at $n=5$, and approaching the infinite value 4 ✓.

Align the powers before subtracting — mis-indexing by one term is the standard AGP error.

Engine 2: the method of differences (telescoping)

KEY RESULT

Telescoping: find f with term = f(k) − f(k+1)

If the general term equals $f(k)-f(k+1)$, then $\sum_{k=1}^{n}=f(1)-f(n+1)$: only the ends survive. The standard split, from partial fractions:

$$\frac{1}{k(k+m)}=\frac{1}{m}\left(\frac{1}{k}-\frac{1}{k+m}\right)$$

With $m=1$: $\sum_{1}^{n}\frac{1}{k(k+1)}=\frac{n}{n+1}$. With $m=2$ the first **two** and last **two** terms survive: $\sum_{k=1}^{5}\frac{1}{k(k+2)}=\frac{1}{2}\left(1+\frac{1}{2}-\frac{1}{6}-\frac{1}{7}\right)=\frac{25}{42}$.

WORKED EXAMPLE

Worked — recognising telescopes in disguise

- $\sum k\cdot k!=(n+1)!-1$, since $k\cdot k!=(k+1)!-k!$.
- $\sum\frac{k}{(k+1)!}=1-\frac{1}{(n+1)!}$, since $\frac{k}{(k+1)!}=\frac{1}{k!}-\frac{1}{(k+1)!}$.
- $\sum_{k=1}^{n}\frac{1}{k(k+1)(k+2)}=\frac{n(n+3)}{4(n+1)(n+2)}$ via $\frac{1}{k(k+1)(k+2)}=\frac{1}{2}\left(\frac{1}{k(k+1)}-\frac{1}{(k+1)(k+2)}\right)$ — a difference of differences.

COMMON TRAP

Common traps — summation engines

- Forgetting the $\frac{1}{m}$ factor in the split.
- Infinite AGP/telescoping sums quoted without the convergence condition ($|r| < 1$; for telescopes, $f(n+1)$ must have a limit).
- With $m \geq 2$, cancelling as if only one term survives at each end.
- Shift-and-subtract with misaligned powers (write both sums out in full, aligned by power of r).

LESSON 05 / 05

The question shapes

KEY RESULT

Shape recognition for Part I problems

1. **Identify the progression from data:** given a_m, a_n in an AP, $d = \frac{a_n - a_m}{n-m}$; in a GP, $r^{n-m} = \frac{a_n}{a_m}$. Given three terms with a condition, use the symmetric parametrisations $(a-d, a, a+d$ / $\frac{a}{r}, a, ar)$.
2. S_n **given,** a_n **wanted:** $a_n = S_n - S_{n-1}$ ($n \geq 2$), $a_1 = S_1$. Watch for S_n formulas whose implied a_1 disagrees — then the sequence is only eventually an AP.
3. **AP meets GP:** the same three numbers in both progressions force them equal (AM = GM with the middle term common). Terms of an AP exponentiated give a GP, and vice versa via logs.
4. **Sum with polynomial general term:** split into $\sum k, \sum k^2, \sum k^3$. Polynomial × power: AGP. Rational with AP factors: telescope.
5. **Inequality/extremum:** AM–GM with constant product and attainable equality; or n^2-type identities for sums of odds.

WORKED EXAMPLE

Worked — mixed set

AP data: an AP has $a_5 = 17$, $a_9 = 29$; $d = \frac{29-17}{9-5} = 3$, $a_1 = 17 - 4 \cdot 3 = 5$, so $a_n = 3n + 2$ and $S_{20} = 10(10 + 57) = 670$? — compute carefully: $S_{20} = \frac{20}{2}(2 \cdot 5 + 19 \cdot 3) = 10 \cdot 67 = 670$ ✓.

Recover terms: $S_n = 3n^2 + 5n$ gives $a_n = S_n - S_{n-1} = 6n + 2$ ($n \geq 2$) and $a_1 = S_1 = 8 = 6 + 2$ consistent, so it is the AP $8, 14, 20, \ldots$ with $d = 6$ ✓.

Identify: is $2, 3, 5, 9, 17, \ldots$ anything? Differences $1, 2, 4, 8$ form a GP — so $a_n = 2^{n-1} + 1$ (each term is one more than a GP term; handle by splitting, sheet Olympiad-02's fixed-point idea in embryo).

KEY RESULT

Recap — the chapter on one card

AP: $a_n = a + (n-1)d$, $S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + \ell)$; $2b = a + c$; means divisor $k+1$. GP: $a_n = ar^{n-1}$, $S_n = \frac{a(r^n - 1)}{r - 1}$, $S_\infty = \frac{a}{1-r}$ ($|r| < 1$); $b^2 = ac$. AM $\geq$ GM for positives, equality iff equal. $\sum k = \frac{n(n+1)}{2}$, $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$, $\sum k^3 = \left(\frac{n(n+1)}{2}\right)^2$. AGP: shift-subtract; $\sum kx^k = \frac{x}{(1-x)^2}$. Telescope: $\frac{1}{k(k+m)} = \frac{1}{m}\left(\frac{1}{k} - \frac{1}{k+m}\right)$, ends survive.

COMMON TRAP

The chapter checklist (mistakes that cost marks)

1. Means divisor k instead of $k+1$. 2. S_∞ with $|r| \geq 1$. 3. Mixed $\frac{r^n - 1}{r-1}$ / $\frac{1-r^n}{1-r}$ signs. 4. Missing $\frac{1}{m}$ in the telescope split. 5. AGP powers misaligned. 6. AM–GM on negatives or with unattainable equality. 7. Assuming AP/GP without checking both difference and ratio. 8. $S_n = \frac{n}{2}(a + \ell)$ with wrong term count. 9. $\sum k^3$ without the square. 10. Recurrence/telescoping answer not checked at the first term.

INTUITION

Forward map

Harmonic progressions and the full $A \geq G \geq H$ chain, recurrences with characteristic equations, generating functions, telescoping products and Wallis, Stolz–Cesàro: Part II. The through-line of the whole topic: *find the representation in which the sum collapses* — pairing (AP), shifting (GP/AGP), or differencing (telescopes).

LESSON 01 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Harmonic progressions are **not** in the official JEE Advanced syllabus (which lists AP, GP, AM/GM and the power sums only). They appear in IOQM/RMO-style material and older JEE Main contexts. The technique — reciprocate, solve the AP, reciprocate back — is the transferable part.

HP via reciprocals

KEY RESULT

Definitions and the only method

A **harmonic progression** is a sequence whose reciprocals form an AP. There is no closed formula for its n-th term or sum comparable to AP/GP: every HP problem is routed through the reciprocal AP.

The **harmonic mean** of $a, b > 0$ is $H = \frac{2ab}{a+b} = \frac{2}{\frac{1}{a}+\frac{1}{b}}$ — defined by $\frac{1}{a}, \frac{1}{H}, \frac{1}{b}$ in AP.

For n numbers, $H = \frac{n}{\sum 1/a_i}$.

WORKED EXAMPLE

Worked — HP basics

HM of 2 and 4: $H = \frac{2 \cdot 2 \cdot 4}{6} = \frac{8}{3}$. The HP $1, \frac{1}{2}, \frac{1}{3}, \dots$ has reciprocals $1, 2, 3, \dots$ (an AP), so its 4-th term is $\frac{1}{4}$. General HP term: if the reciprocal AP has first term $\frac{1}{a}$ and difference d', the n-th HP term is $\frac{1}{\frac{1}{a}+(n-1)d'}$.

KEY RESULT

The chain A ≥ G ≥ H and the identity $G^2 = AH$

For two positive numbers, $A = \frac{a+b}{2}$, $G = \sqrt{ab}$, $H = \frac{2ab}{a+b}$ satisfy $A \geq G \geq H$, and moreover

$$G^2 = AH$$

since $\frac{a+b}{2} \cdot \frac{2ab}{a+b} = ab$. So A, G, H are in GP. Equality throughout iff $a = b$. For n positives, $A \geq G \geq H$ still holds (apply AM-GM to the numbers and to their reciprocals).

STANDARD APPLICATION

Standard applications

Average speed over equal distances at speeds u, v is the **harmonic** mean $\frac{2uv}{u+v}$, not the arithmetic mean — the canonical HP-in-disguise. Work-rate and parallel-resistance formulas are the same shape: whenever the reciprocals add, H appears.

COMMON TRAP

Common traps — HP

- Inventing an "S_n formula" for HP by analogy with AP/GP: none exists in elementary closed form; sums route through $H_n = \sum \frac{1}{k}$ estimates only.
- H for numbers of mixed sign: the mean framework assumes positives.
- Confusing "insert harmonic means" (reciprocate, insert AMs, reciprocate back) with inserting AMs directly.

LESSON 02 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Linear recurrences and generating functions sit outside the official AP/GP syllabus lines; they are the standard Olympiad machinery for sequences defined inductively.

First- and second-order recurrences

SHORT DERIVATION

Fixed-point shift for $a_n = p\cdot a_{n-1} + q$

Seek L with $L = pL + q$, i.e. $L = \frac{q}{1-p}$ $(p \neq 1)$. Then $a_n - L = p(a_{n-1} - L)$: a GP in disguise, so $a_n = L + (a_1 - L)p^{n-1}$. Worked: $a_n = 2a_{n-1} + 1$, $a_1 = 1$: $L = -1$, $a_n = 2^n - 1$ ✓ $(1, 3, 7, 15, \ldots)$. If $p = 1$ it is an AP instead.

KEY RESULT

Characteristic equation for $a_n = p\cdot a_{n-1} + q\cdot a_{n-2}$

Try $a_n = r^n$: the **characteristic equation** is $r^2 - pr - q = 0$. Distinct roots $r_1 \neq r_2$: $a_n = Ar_1^n + Br_2^n$; repeated root r: $a_n = (A + Bn)r^n$. Constants from two initial values. Always verify the closed form against the first computed terms.

Generating functions

KEY RESULT

The clothesline: operations on $G(x)$ are operations on the sequence

$G(x) = \sum_{n \geq 0} a_n x^n$. Multiply by x: shift right. Multiply by $\frac{1}{1-x}$: replace a_n by its partial sums. Differentiate: multiply a_n by n. $G(-x)$: alternate signs. Recurrences become algebraic equations for $G(x)$; solving and expanding reads off a_n.

SHORT DERIVATION

Fibonacci: $G(x)$ to Binet in three moves

With $F_0 = 0$, $F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$: comparing coefficients of x^n $(n \geq 2)$ in $F(x)$ and $xF(x) + x^2F(x) + x$ gives

$$F(x) = \frac{x}{1 - x - x^2} = \frac{x}{(1 - \varphi x)(1 - \psi x)}, \qquad \varphi, \psi = \frac{1 \pm \sqrt{5}}{2}$$

Partial fractions: $F(x) = \frac{1}{\sqrt{5}}\left(\frac{1}{1-\varphi x} - \frac{1}{1-\psi x}\right)$; expanding both geometric series,

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$$

Checks: $F_1 = \frac{2.236068}{2.236068} = 1$ ✓; $F_7 = 13$ ✓; $F_{10} = 55$ ✓. This is Binet with no induction — pure algebra (and it matches the quadratic route of chapter 6, Olympiad sheet 01).

COMMON TRAP

Common traps — recurrences and generating functions

- Characteristic equation sign: for $a_n = pa_{n-1} + qa_{n-2}$ it is $r^2 - pr - q = 0$, not $r^2 + pr + q$.
- Repeated-root case forgotten: $(A + Bn)r^n$, not Ar^n.
- Closed form not checked at $n = 1, 2$: index-shift errors are invisible otherwise.
- Generating-function manipulations treated as convergent series: for coefficient extraction they are formal power series; convergence is a separate question.

LESSON 03 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Multi-factor telescopes, infinite products, discrete l'Hôpital and growth-rate answers are Olympiad/analysis-flavoured extensions of the Part I summation engines.

Products telescope too

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

Products telescope too

SHORT DERIVATION

The product $\prod(1 - 1/k^2)$

Since $\log\prod a_k = \sum \log a_k$, products collapse by the same cancellation as sums. The unlocking identity is the difference of squares: $1 - \frac{1}{k^2} = \frac{k-1}{k}\cdot\frac{k+1}{k}$. Hence

$$\prod_{k=2}^{n}\left(1-\tfrac{1}{k^2}\right) = \prod_{k=2}^{n}\tfrac{k-1}{k}\cdot\prod_{k=2}^{n}\tfrac{k+1}{k} = \tfrac{1}{n}\cdot\tfrac{n+1}{2} = \tfrac{n+1}{2n} \to \tfrac{1}{2}$$

Check at $n=4$: $\frac{3}{4}\cdot\frac{8}{9}\cdot\frac{15}{16} = \frac{5}{8} = \frac{4+1}{2\cdot 4}$ ✓. The same split in $\prod\frac{4k^2}{4k^2-1} = \prod\frac{2k}{2k-1}\cdot\frac{2k}{2k+1}$ gives the finite closed form $\frac{2^{4n}(n!)^4}{(2n+1)((2n)!)^2}$, whose limit is **Wallis's product** $\frac{\pi}{2}$.

Stolz–Cesàro: l'Hôpital for sequences

KEY RESULT

The discrete l'Hôpital

If b_n is strictly increasing and unbounded, then $\lim\frac{a_n}{b_n} = \lim\frac{a_{n+1}-a_n}{b_{n+1}-b_n}$ whenever the latter limit exists. No differentiability needed.

Worked: $\lim\frac{H_n}{\ln n}$ with $H_n = \sum_1^n \frac{1}{k}$: the difference ratio is $\frac{1/(n+1)}{\ln(1+1/n)} \to 1$, so $\frac{H_n}{\ln n} \to 1$. Numerically $1.1264, 1.0836, 1.0627, 1.0501$ at $n = 10^2, \ldots, 10^5$ — approaching 1 slowly, at rate $\sim \frac{\gamma}{\ln n}$ with γ Euler's constant.

Asymptotics: when the honest answer is a growth rate

WORKED EXAMPLE

Worked — $a_{n+1} = a_n + 1/a_n$ grows like $\sqrt{2n}$

$a_1 = 1$, $a_{n+1} = a_n + \frac{1}{a_n}$. Square: $a_{n+1}^2 = a_n^2 + 2 + \frac{1}{a_n^2}$, telescoping to $a_n^2 = 2(n-1) + \sum_{k=1}^{n-1}\frac{1}{a_k^2} + 1$. Since $a_k^2 \approx 2k$, the terms $\frac{1}{a_k^2} \approx \frac{1}{2k}$, so the sum grows only like $\frac{1}{2}\ln n$; hence $a_n^2 = 2n + O(\log n)$ and $a_n \sim \sqrt{2n}$. (Note: the remainder sum does **not** converge to a constant — it diverges logarithmically; the growth-rate conclusion is unaffected.) Numerics: $a_{10} = 4.5699$ vs $\sqrt{20} = 4.4721$, ratio $\to 1$ ✓. The method: **transform until it telescopes, then estimate the remainder.**

COMMON TRAP

Common traps — advanced summation

- Infinite product "cancellation" without checking convergence to a **nonzero** limit.
- Stolz–Cesàro applied with non-monotone denominator, or in the converse direction (the difference-limit may fail while the ratio-limit exists).
- Asymptotic claims stated as equalities: $a_n \sim \sqrt{2n}$ means ratio $\to 1$, not $a_n = \sqrt{2n} + o(1)$.
- Telescoping arctan-type sums (e.g. $\sum\arctan\frac{1}{k^2+k+1} = \frac{\pi}{4} - \arctan\frac{1}{n+1}$) need branch care: each difference must land in the principal range (chapter 4 of this series).